Explaining BiomedCLIP with Weighted Banzhaf Interactions supported by Tree-Gram Parsing

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Abstract

Vision-Language Models (VLMs) in the medical domain require explanations that align with precise clinical terminology. We introduce an extension to standard FIxLIP attribution, incorporating linguistic dependency parsing trees to define players in the Banzhaf interaction game. By grouping tokens we address the raw tokens problem and provide semantically coherent cross-modal explanations of BiomedCLIP on the radiology dataset. Code and results are available on: https://github.com/kubarrr/ParsingFIxLIP

1. Introduction

Vision–Language Models (VLMs) such as BiomedCLIP (Zhang et al., 2023) show strong performance on medical image–text tasks, but their deployment in clinical settings requires both faithful and interpretable explanations. Existing methods, like FIxLIP (Baniecki et al., 2025), operate at subword tokens produced by modern tokenizers, resulting in semantically incoherent explanations split across multiple tokens. In this work, we extend the FIxLIP framework by introducing Tree-Gram Parsing idea (Sima'an, 2000), a linguistically informed strategy for defining explanation players, using it to group tokens into semantically meaningful units, and then produce cross-modal explanations for BiomedCLIP.

Our contributions can be summarized as follows: we adapt the FIxLIP interaction-based explanation framework to BiomedCLIP and radiology data, we propose Tree-Gram Parsing, which uses NLP parsing trees to mitigate explanation fragmentation by operating on semantically coherent textual units, and we empirically evaluate multiple grouping strategies on the ROCOv2 dataset (Rückert et al., 2024), showing that while quantitative fidelity metrics are just slightly higher in comparison to token-level baselines, parsing-based explanations provide clearer and more clinically interpretable qualitative insights.

2. Methodology

2.1 FIxLIP

FIxLIP is a framework designed for vision–language models such as CLIP, aimed at revealing how predictions arise from image-text partial interactions. It decomposes the model's output into main effects and pairwise interactions, making it possible to identify *how* parts of the image or text influence each other. The scores are computed using Banzhaf interaction

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values, quantifying the average marginal contribution of features and interactions. Naive FIxLIP masking is based on random hiding of each token separately.

2.2 Tree-gram parsing

The main improvement we propose is applying Tree-Gram Parsing methods with spaCy tokenizer (Honnibal et al., 2020) (module `en_core_web_sm`). We do not mask tokens separately, but grouped by players, who can get tokens variously: `raw` tokenizer splitting (like FIxLIP), splitting by `words`, grouping noun-related words into `noun_chunks` or `smart_depth`, where syntactically meaningful words are grouped with their modifiers. Details in the Section 6.1

3. Quantitative Experiments

We evaluate the proposed parsing strategies on a subset of the ROCOv2 dataset following the evaluation protocol of (Baniecki et al., 2025). Since explanation quality strongly depends on report length, we stratify the analysis by caption length and focus the main quantitative results on short radiology captions. More details in the Section 6.2.

3.1 Faithfulness: Spearman Correlation and Adjusted R^2

The faithfulness is measured by Spearman's Rank Correlation and Adjusted R^2 **to check if Tree-Gram Parsing increases it**. Spearman's correlation assesses the general capability of the explanation method to recover the ranking of similarity scores for uniformly masked inputs compared to the original model predictions. Adjusted R^2 is used to evaluate the precise goodness-of-fit. It accounts for the varying number of players resulting from different parsing strategies. In terms of Spearman correlation, `noun_chunks` and `smart_depth` demonstrate the highest stability with a median score close to 0.9. This disparity is more visible in the Adjusted R^2 metric, where both `smart_depth` and `noun_chunks` achieve median values above 0.7.

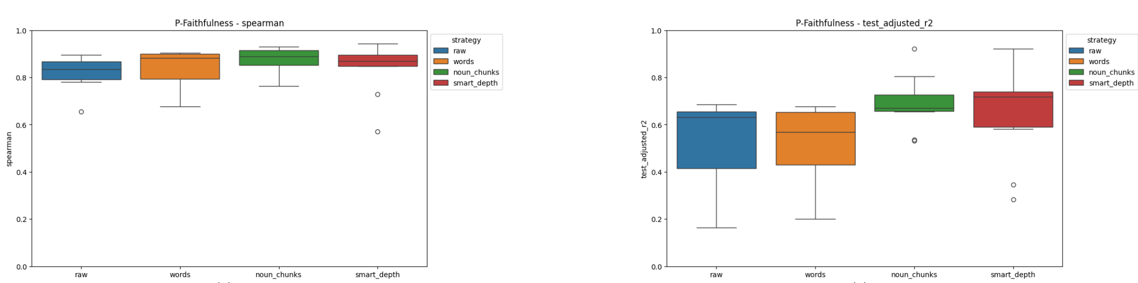

Figure 1: Faithfulness evaluation on 10 short captions samples. **Left:** Spearman Rank Correlation. **Right:** Adjusted R^2. Noun Chunks and Smart Depth **consistently outperform the token-level baselines**

3.2 Insertion and Deletion Curves

We further assess the quality of the explanations, **to check if Tree-Gram Parsing improves them**, using Insertion and Deletion curves, quantified by the Area Between Curves (AID). These metrics measure how rapidly the model's confidence changes as important features are added to or removed from the input. On Figure 2 semantic grouping strategies achieve consistently higher AID scores compared to token-level baselines. `Noun_chunks` yields the highest AID (0.87 ± 0.20), followed closely by `smart_depth` (0.86 ± 0.26), **significantly outperforming** the `raw` strategy (0.77 ± 0.24).

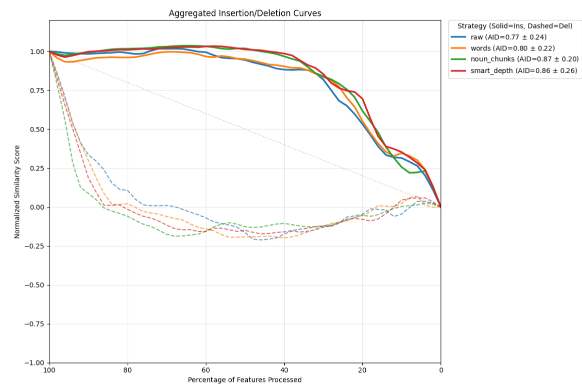

Figure 2: Insertion and Deletion curves for 10 short captions samples.

4. Qualitative Experiments

4.1 Parsing enables to understand full medical concepts

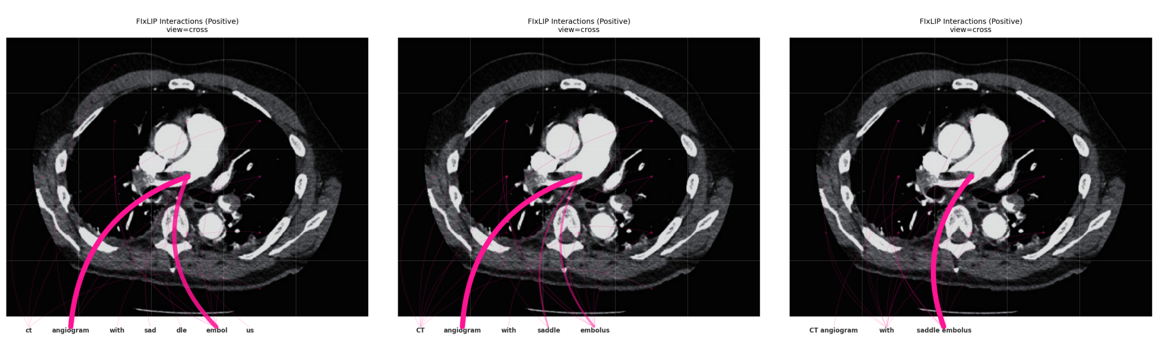

Figure 3: Qualitative comparison of explanations. **Left:** Raw - Concept fragmented, noisy explanation. **Middle:** Words - Subwords merged into words. **Right:** Smart Depth - "saddle embolus" as one concept, a clean explanation.

To see if Tree-gram parsing challenges the key limitation of standard VLE explanations — the fragmentation of medical concepts caused by tokenization. Figure 3 illustrates the progression of interpretability across different parsing strategies for the caption "CT angiogram with saddle embolus". In the `raw` strategy the medical term "saddle embolus" is split into arbitrary tokens (e.g., "sad", "dle", "embol", "us"). The explanation is noisy, with importance distributed across meaningless fragments and significant weight given to the general modality "angiogram" rather than the specific pathology "saddle embolus". The `words` strategy merge subwords into recognizable terms. However, it split similarity score treating "saddle" and "embolus" as separate players, still highlighting general term "angiogram". In contrast, our proposed semantic parsing strategy, `smart_depth` successfully groups these tokens into a coherent clinical concept. The parser identifies "saddle embolus" as a single entity. Consequently, the explanation concentrates mostly on this unified concept and results in best quantitative metrics too. We attach other examples in the Section 6.4.

4.2 Image transformation test

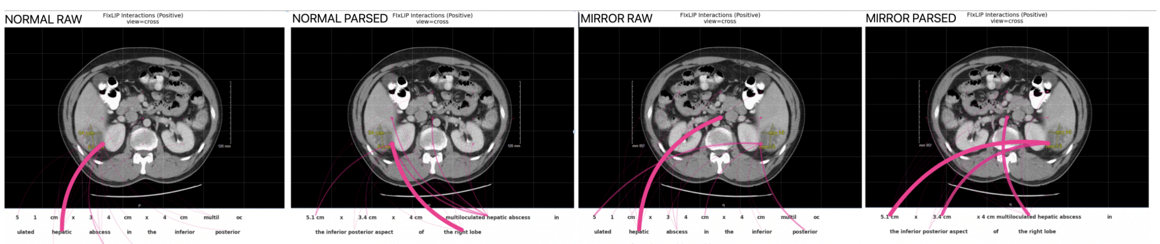

Figure 4: Image transformation comparison. **First:** Normal raw, **Second:** Normal Smart Depth, **Third:** Mirrored raw, **Fourth:** Mirrored Noun Chunks

Is parsing on BiomedCLIP robust to rotations and image mirroring? Normal image has a similarity 0.4587, mirror — 0.3061. This is probably caused by training on the medical images took always from the same position, thus the model does not recognize the concept of the right lobe in the mirrored image and the main contributor are the numbers. Grouping tokens strengthens "the right lobe" score for normal view. **The model seems to be robust to transformations in the numbers' and symbols aspect but not in the body parts' view aspect**. We provide more explanations in the Section 6.5.

5. Conclusions

Adding Tree-Gram Parsing to the FIxLIP algorithm allows us to understand more about how tokens influence the model results and find out the synergy of the grouped tokens. Quantitative evaluation shows that our parsing-based approach achieves performance higher than the raw token-based FIxLIP. While computational constraints limited the scale of our quantitative experiments, qualitative analysis demonstrates that our dependency-based grouping significantly mitigates the fragmentation issues inherent in the baseline method, offering more intuitive and clinically relevant insights into the model's decision-making process.

References

Hubert Baniecki, Maximilian Muschalik, Fabian Fumagalli, Barbara Hammer, Eyke Hüllermeier, and Przemysław Biecek. Explaining similarity in vision-language encoders with weighted banzhaf interactions. *arXiv preprint arXiv:2508.05430*, 2025.

Matthew Honnibal, Ines Montani, Sofie Van Landeghem, and Adriane Boyd. spacy: Industrial-strength natural language processing in python, 2020. URL `https://spacy.io`.

Johannes Rückert, Louise Bloch, Raphael Brüngel, Ahmad Idrissi-Yaghir, Henning Schäfer, Cynthia S. Schmidt, Sven Koitka, Obioma Pelka, Asma Ben Abacha, Alba G. Seco de Herrera, Henning Müller, Peter A. Horn, Felix Nensa, and Christoph M. Friedrich. Rocov2: Radiology objects in context version 2, an updated multimodal image dataset, 2024. URL `https://arxiv.org/abs/2405.10004`.

K. Sima'an. Tree-gram parsing: Lexical dependencies and structural relations. In *Proceedings of the 38th Annual Meeting of the Association for Computational Linguistics*, pages 37–44, Hong Kong, 2000. Association for Computational Linguistics. doi: 10.3115/1075218.1075226.

Sheng Zhang, Yanbo Xu, Naoto Usuyama, Hanwen Xu, Jaspreet Bagga, Robert Tinn, Sam Preston, Rajesh Rao, Mu Wei, Naveen Valluri, Cliff Wong, Andrea Tupini, Yu Wang, Matt Mazzola, Swadheen Shukla, Lars Liden, Jianfeng Gao, Angela Crabtree, Brian Piening, Carlo Bifulco, Matthew P. Lungren, Tristan Naumann, Sheng Wang, and Hoifung Poon. Biomedclip: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs, 2023. URL `https://arxiv.org/abs/2303.00915`.

6. Appendix

6.1 Parsing strategies

Strategy	Grouping principle	Linguistic basis	Granularity	Typical use case
`raw`	One player per tokenizer subtoken	Tokenizer (BPE / WordPiece)	Very fine	Exact alignment with model inputs and low-level attribution
`words`	One player per full word, merging subtokens	spaCy tokenization	Fine	Readable word-level explanations without subword noise
`noun_chunks`	Multi-word noun phrases grouped, remaining tokens standalone	Constituency (noun phrases)	Medium	Entity- and object-centric interpretation
`smart_depth`	Syntactic heads grouped with modifiers (e.g. negation, adjectives, auxiliaries)	Dependency parsing	Medium-coarse	Semantically coherent action or state representations

Table 1: Token grouping strategies implemented in `TextParser`

Strategy	Players (name → token indices)
`raw`	`viral → [1], pneumonia → [2], is → [3], not → [4], ruled → [5], out → [6]`
`words`	`Viral → [1], pneumonia → [2], is → [3], not → [4], ruled → [5], out → [6]`
`noun_chunks`	`Viral pneumonia → [1,2], is → [3], not → [4], ruled → [5], out → [6]`
`smart_depth`	`Viral pneumonia → [1,2], is not ruled out → [3,4,5,6]`

Table 2: Examples of token grouping produced by different parsing strategies for the sentence “Viral pneumonia is not ruled out.”

6.2 Quantitative experiments' technical details

We evaluate the proposed parsing strategies on a subset of the ROCOv2 dataset using the evaluation protocol FIxLIP authors (Baniecki et al., 2025). To assess out-of-distribution generalization, we employ 1,000 random binary masks per input, with each feature independently masked with probability $p = 0.5$, consistent with the training setup.

For the approximation setup, we use a sampling budget of up to 50,000 image-text input pairs for the estimation. Consistent with the efficient approximation design in FIxLIP, the explanation basis is restricted to main effects and cross-modal interactions.

We observe that the length of the radiological report has a significant impact on the complexity of the explanation task. In particular, long captions, which contain dense spatial and numerical information, are substantially harder to approximate. Under the fixed computational budget, even the training R^2 remains low, indicating underfitting due to the increased combinatorial complexity of the explanation space.

Consequently, we stratify the analysis by caption length, distinguishing between short captions (< 10 words) and long captions (> 30 words). The quantitative analysis in the main body of the paper focuses on short captions, while the results for long captions, including additional error analysis and discussion, are provided in the Section 6.3.

6.3 Quantitative metrics for long captions samples

We extend our quantitative evaluation to the challenging subset of long captions (> 30 words). In this setting, the interaction space expands drastically. Combining approximately sometimes even 60 text tokens with 16 image patches results in a high-dimensional feature space populated largely by functional words and background noise. We employed a sampling budget of $N = 50{,}000$ perturbations. The results suggest that this budget might be insufficient for the token-level baseline to stabilize the signal-to-noise ratio in such a sparse feature space.

First, we analyze the model fit in Figure 5. While the Standard R^2 (Left) remains comparable across strategies ($\sim$0.65), a critical divergence appears in the Adjusted R^2 (Right). The `raw` strategy drops to negative values for two samples. This phenomenon likely reflects the curse of dimensionality, as the `raw` method attempts to fit a linear explanation using over 70 variables (tokens + patches). The penalty for this high complexity, combined with the multicollinearity of noisy linguistic tokens, outweighs the marginal gains in fit. In contrast, `smart_depth` effectively performs dimensionality reduction, condensing the input into < 15 semantic concepts. This allows the method to recover a tatistically robust signal.

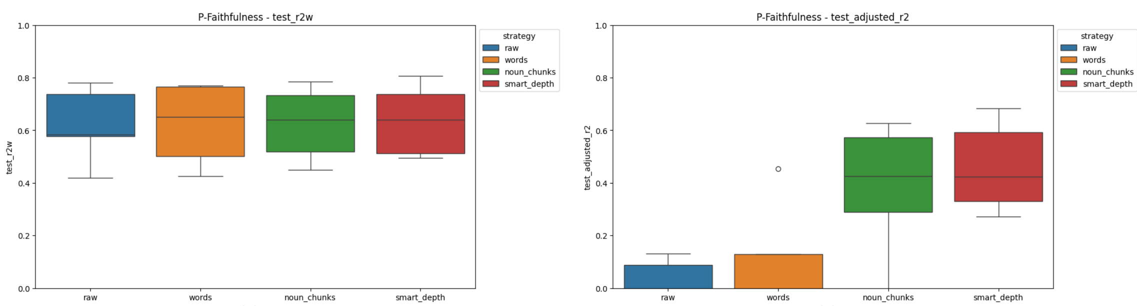

Figure 5: Goodness-of-fit metrics on long captions. **Left:** Standard R^2 shows comparable explanatory potential. **Right:** Adjusted R^2 reveals the inefficiency of the raw strategy, which is penalized for its high feature count, whereas `smart_depth` remains robust due to semantic parsimony.

Figure 6 shows that the `raw` strategy achieves even higher AID scores. This apparent contradiction with the Adjusted R^2 results might be explained by the nature of VLM attention. In long sequences, models may exhibit scattered attention to many non-semantic tokens (”view”, ”shows”, ”the”, single numbers) like visible on Figure 8. Since the `raw` strategy assigns nonzero importance to most tokens — including noise — it appears to be strictly faithful to this scattered behavior. It seems that `smart_depth`, by enforcing semantic grouping, likely filters out this non-clinical noise. This suggests a trade-off where `smart_depth` sacrifices fidelity to the model’s noise artifacts in exchange for the semantic coherence.

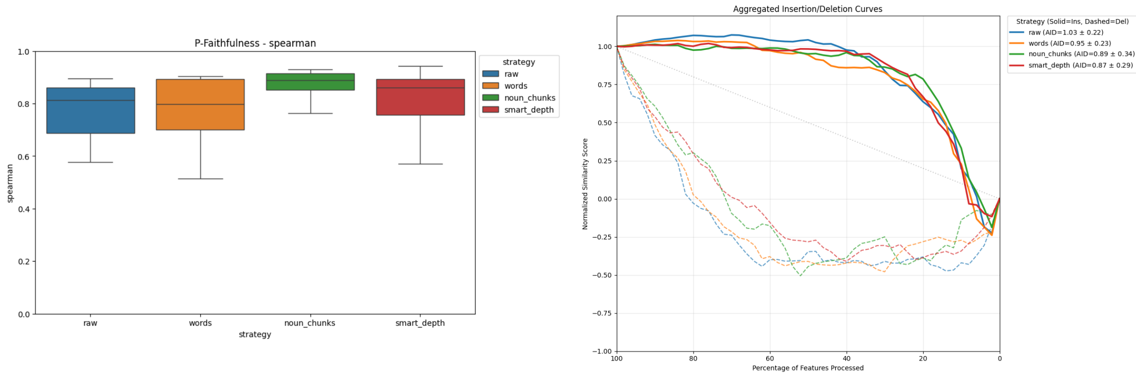

Figure 6: Metrics on long captions. **Left:** Spearman Correlation. **Right:** AID Curves. The higher scores for the long captions, full of adverbs and non-semantic words, suggest raw baseline may track the model's scattered attention to background noise, which semantic grouping strategies intentionally abstract away.

6.4 Other examples of discovering full medical concepts

We further illustrate the capability of our Smart Depth strategy to capture complex, multi-word medical concepts compared to the fragmented baseline.

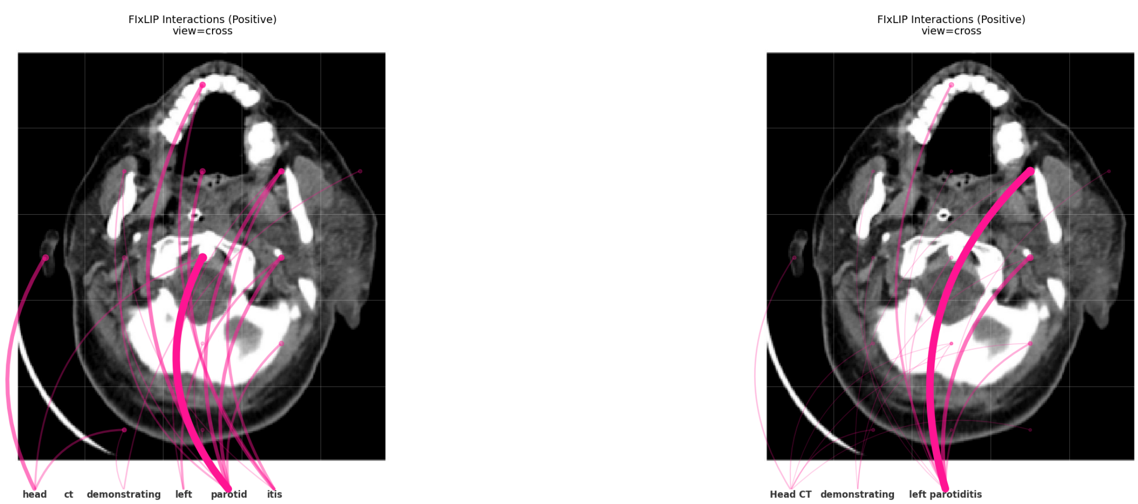

Figure 7: Visual explanation comparison for the caption "Head CT demonstrating left parotiditis". **Left:** Raw - fragmenting the concept into "left", "parotid", and "itis" results in scattered attention. **Right:** Smart Depth - processing "left parotiditis" as a single semantic unit yields a precise explanation. Semantic grouping eliminates the noise seen with fragmented tokens.

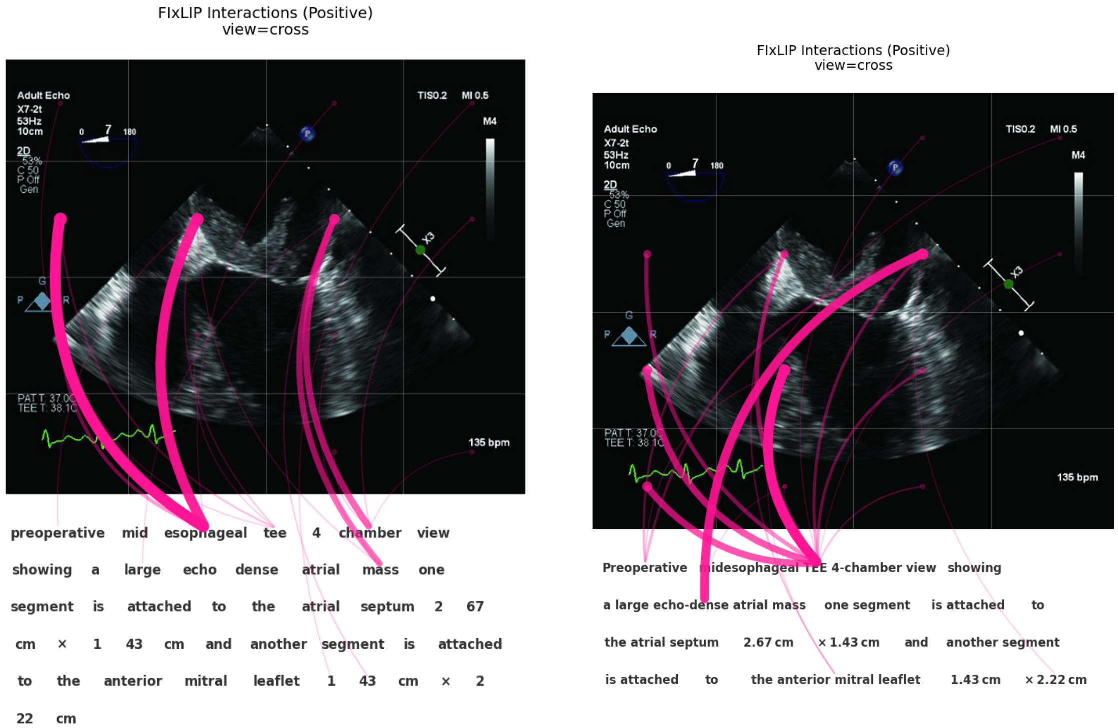

Figure 8: Visual explanation comparison for the long caption and complex phrase "echo-dense atrial mass". **Left:** Raw Strategy - the concept is fragmented into individual tokens (e.g., "echo", "dense", "atrial"), leading to scattered attention. **Right:** Smart Depth - semantic grouping unifies these tokens, assigning importance to this pathology and its anatomical view

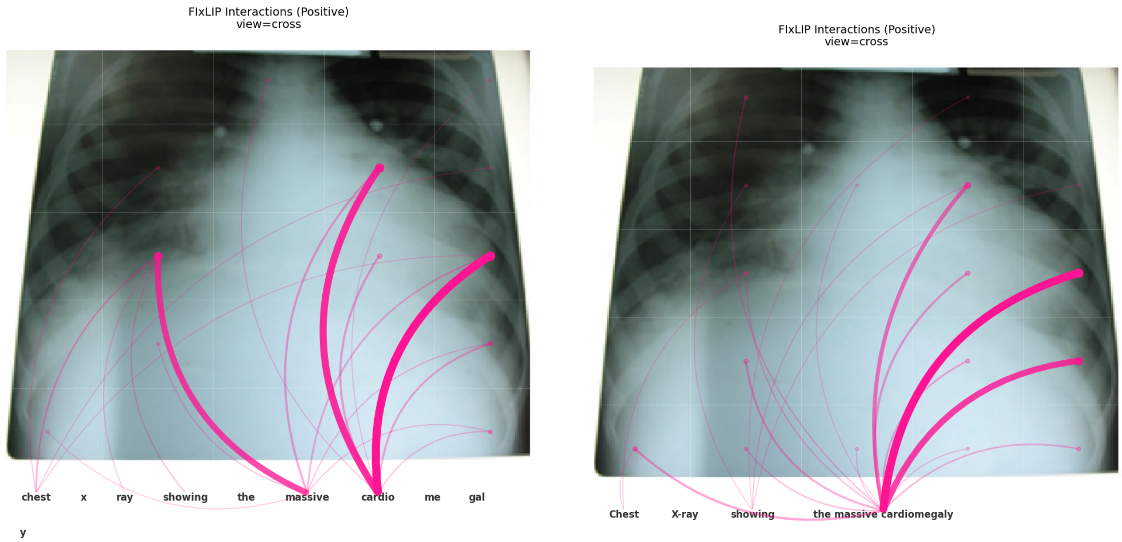

Figure 9: Visual explanation comparison for “massive cardiomegaly”. **Left:** Raw - fragmentation of the medical term generates significant visual noise outside the heart region. **Right:** Smart Depth - processing the entire phrase as a single unit eliminates noise.

6.5 Image transformation test — more information

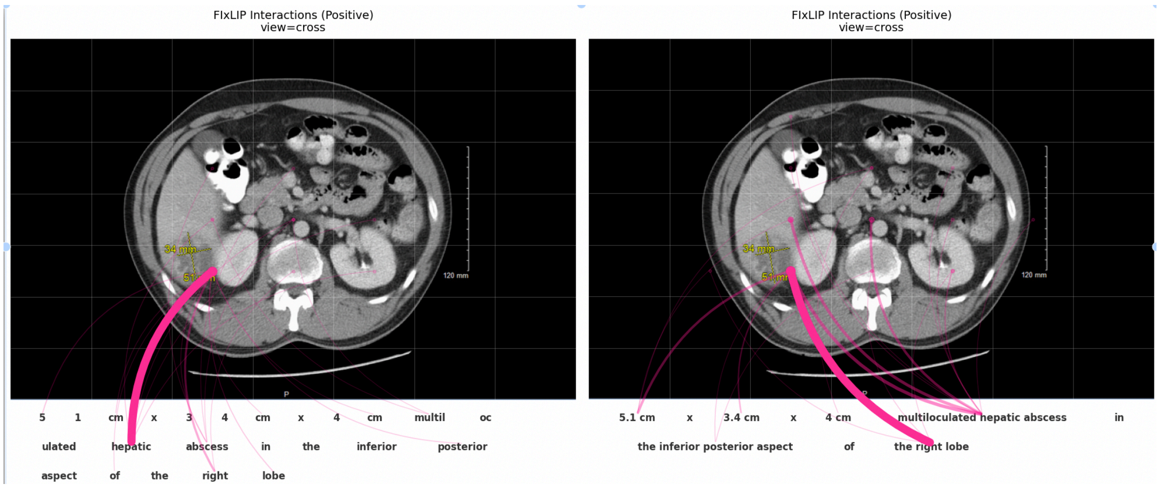

Figure 10: Image transformation comparison. **Left:** Normal raw, **Right:** Normal Smart Depth

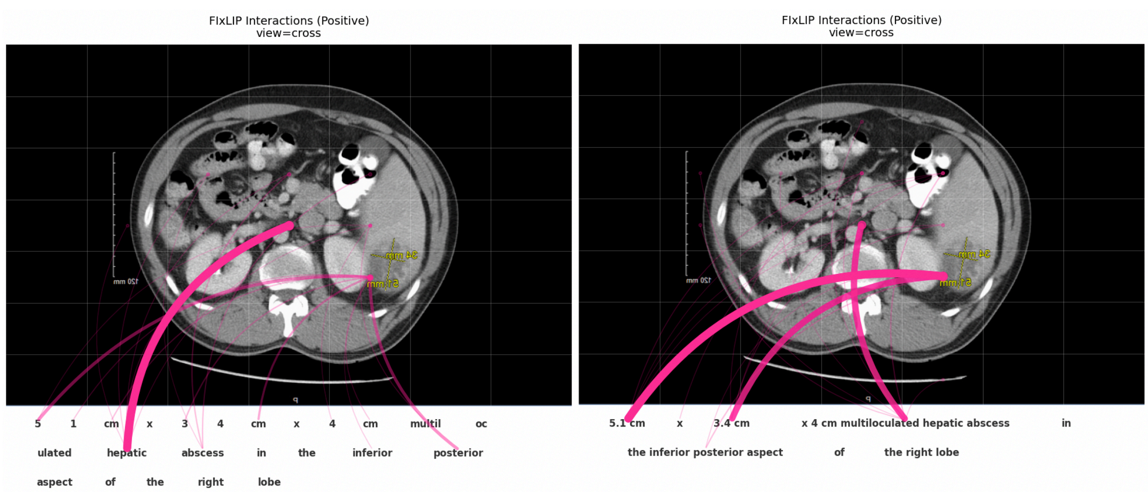

Figure 11: Image transformation comparison. **Left:** Mirrored raw, **Right:** Mirrored Noun Chunks

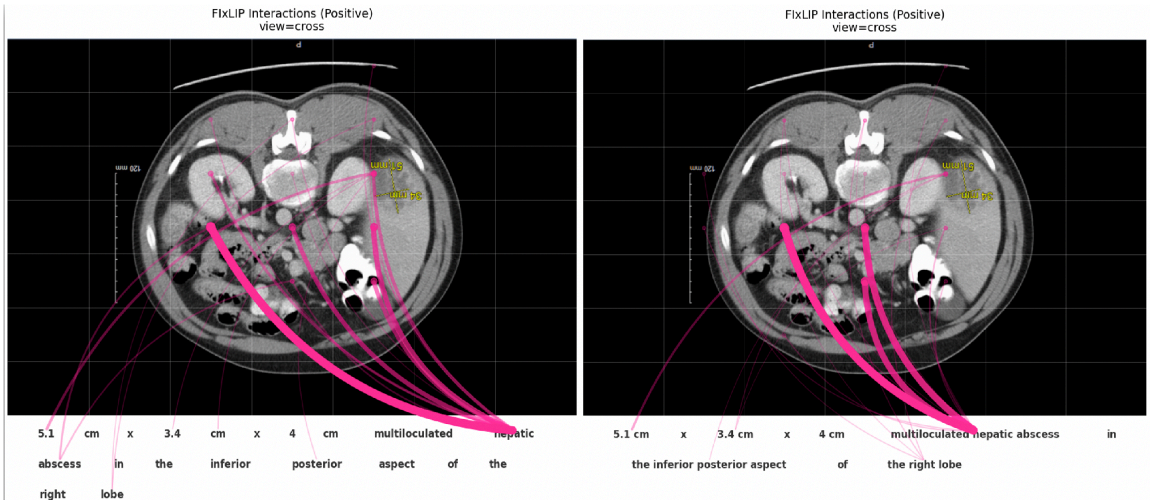

Figure 12: Image transformation comparison. **Left:** Upside down raw, **Right:** Upside down Smart Depth

The model's behavior during rotation upside down (similarity 0.3334) is very weird — it seems that the model is learning "multiloculated hepatic abscess" from areas other than the liver. This may be related to "hepatic" domination seen in the raw, which word regularly appears in the vision context (and can be linked with liver-neighboring organs). What is also worth noticing is that `smart_depth` algorithm not always does generate less players than `noun_chunks`.

6.6 False label test

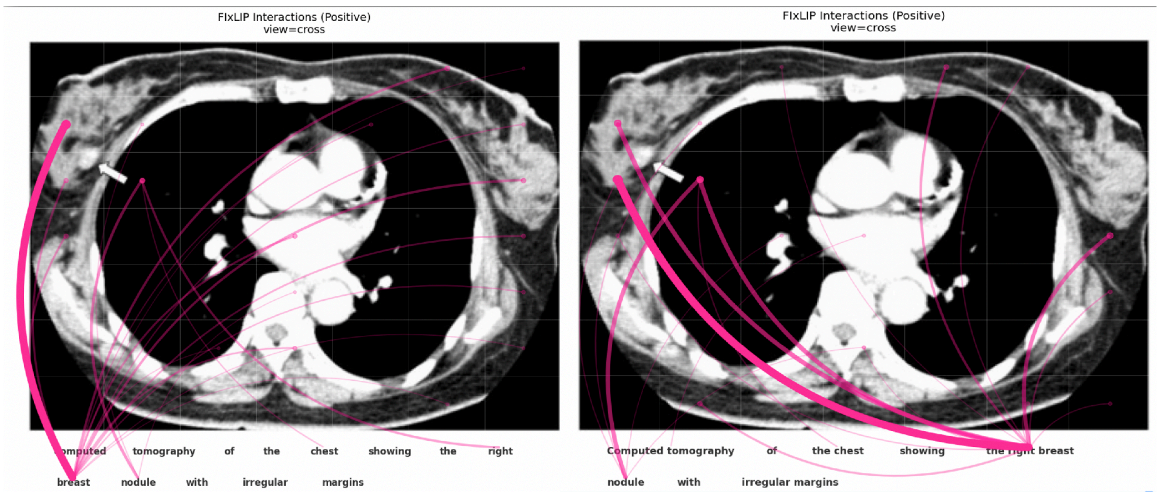

Figure 13: False label test comparison. **Left:** Normal raw, **Right:** Normal Smart Depth

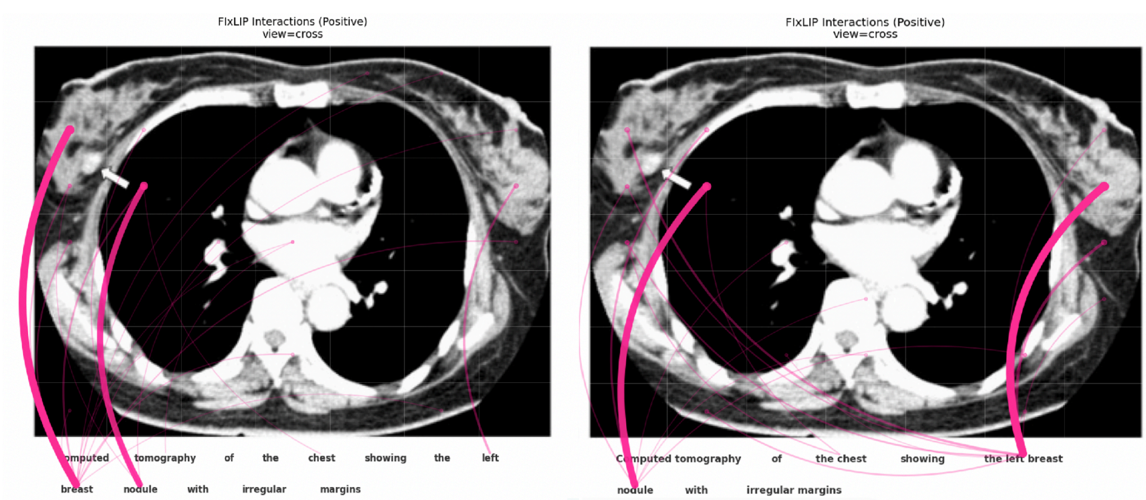

Figure 14: False label test comparison. **Left:** False label raw, **Right:** False label Smart Depth

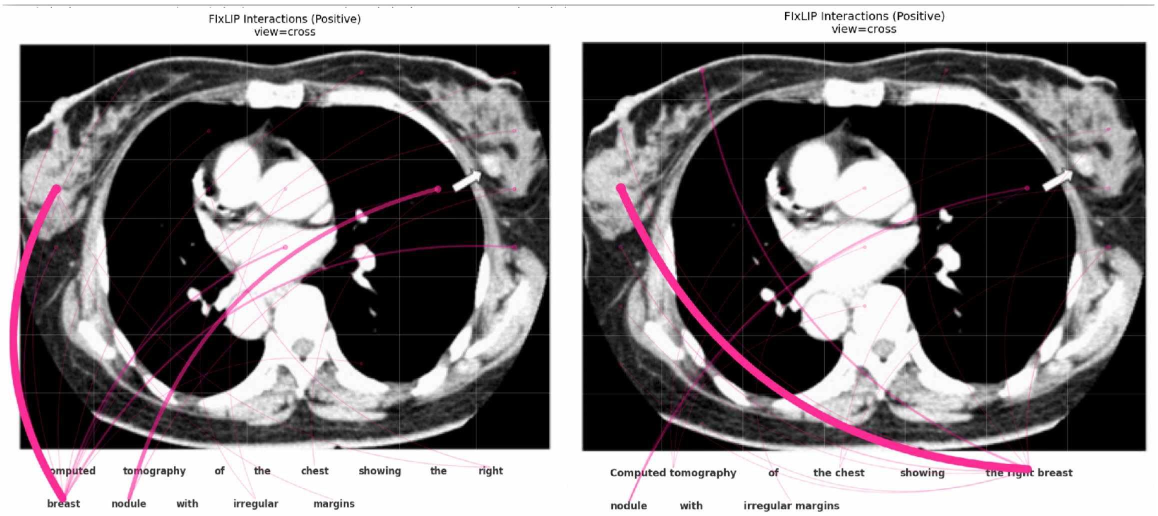

Figure 15: False label test comparison. **Left:** Mirrored raw, **Right:** Mirrored Smart Depth

Can the model detect left/right change in the text? The model can distinguish between the left and the right breast, probably considering its placement in the image. The breast position on the image is meaningful for selecting the correct breast, not the arrow. Connecting the tokens together changes the part of the image indicated by breast — reast appearing always with the correct left/right indicator is linked with the image part correctly, not as the right.

The model fails in recognizing the correct breast in the mirrored image (not surprising, as breasts are quite similar and model is not transformation-robust).

Upsetting in the experiment is that Tree-Gram Parsing algorithm separated breast token from the nodule token (probably due to being separate nouns). It would be interesting to see if the model is more likely to classify such grouped chunk to the left breast (false label) or the nodule (detected correctly on the right breast).

Fun factor: in the image we can see both the nodule and the arrow pointing to it. They are on separate grids, but all the experiments show that the arrow is more meaningful for nodule detection in the image than the nodule itself.

6.7 Difficulties in interpreting numbers or arrows on radiology images

Radiological images frequently contain explicit annotations (arrows, rulers, text) which pose a unique challenge. Our framework connects these straightforward, not interesting from a medical perspective, interactions.

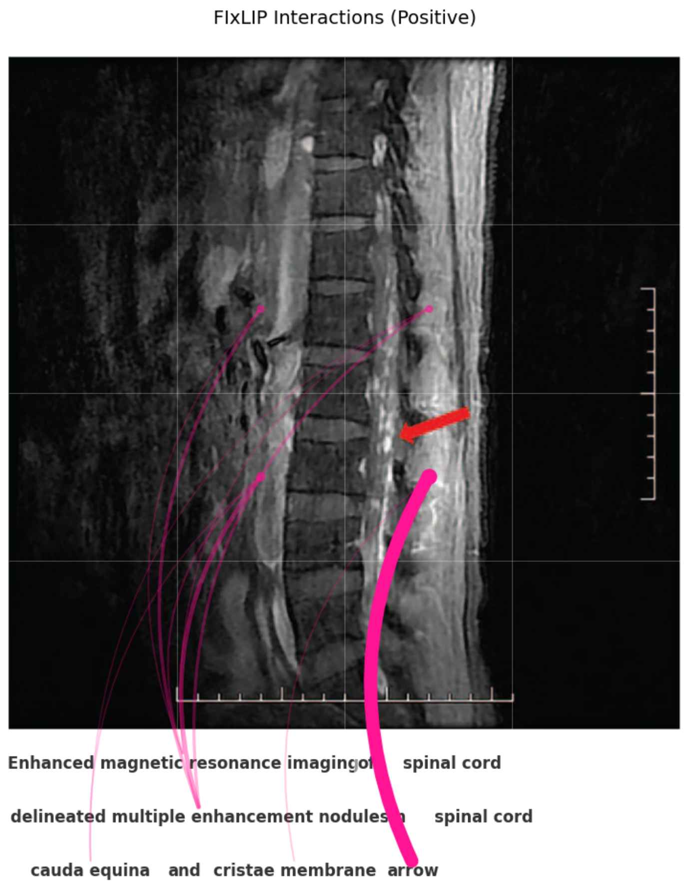

Figure 16: Visual grounding of explicit annotations. The word “arrow” is connected to the red graphical marker pointing towards the spinal pathology. Some other seemingly important relations can be omitted.

6.8 Tree-Gram Parsing on the famous dog with hydrant image

We validated our method on the same image as in FIxLIP (Baniecki et al., 2025): ”black dog next to a yellow hydrant” image. Baseline `raw` strategy identifies “dog” and “hydr” as the tokens with the highest similarity, assigning also smaller weight to “ant”. However, the `smart_depth` strategy significantly enhances faithfulness by aggregating these tokens into semantic words. As shown in the , the `raw` strategy displays mentioned fragmentation. By applying `smart_depth`, we observe that “yellow hydrant” correctly achieves the highest similarity. Conversely, the phrase “black dog” loses similarity strength. The grouping may reveal that the adjective “black” (which contradicts the visual evidence of a white dog) negatively penalizes the positive signal of “dog”. This semantic alignment is quantitatively confirmed by the Adjusted R^2 score, which improves drastically from 0.62 in the `raw` setting to over 0.93 with `smart_depth`.

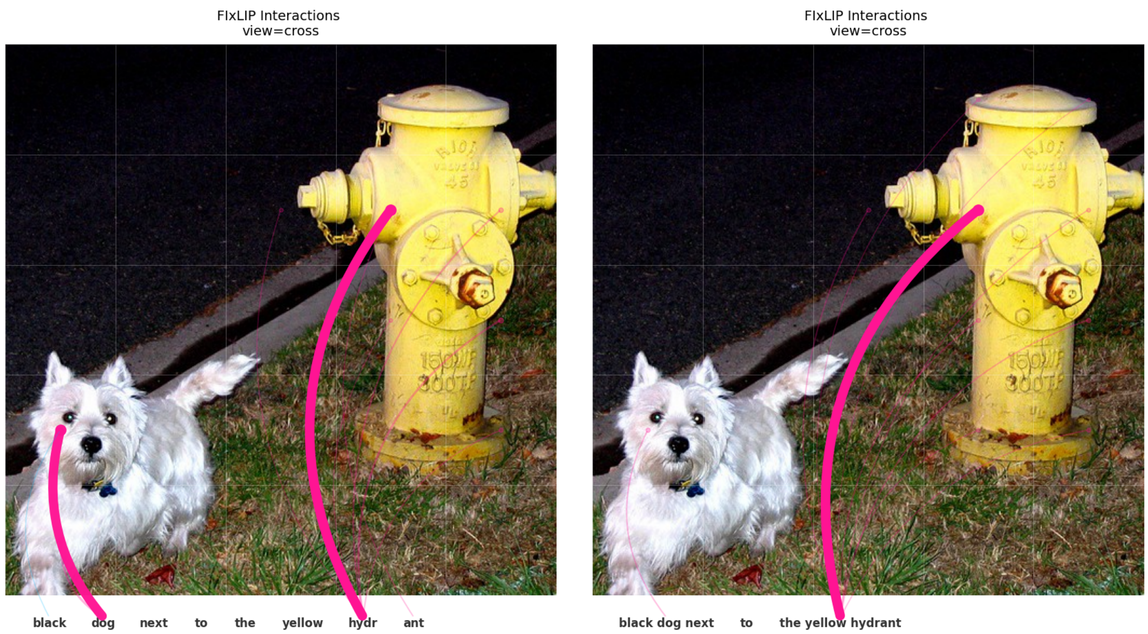

Figure 17: Visual grounding on the prompt ”black dog next to the yellow hydrant”.

To isolate the interaction effects masked by the strong class signal of “dog“ and “hydrant“, we conducted a second experiment with the caption “black animal is running without its friend” as in the (Figure 18). Here, the generic term “animal” has a lower base similarity than “dog”, allowing us to observe the interactions of the adjective more clearly. In the `raw` view, the isolated token “black” exhibits a strong negative influence (blue/negative interactions) on both the yellow hydrant and the white fur of the animal. When processed with `smart_depth`, merging “black” with “animal” creates a unified concept where the negative weight of “black” visibly reduces the overall attribution of the phrase. This confirms that our method successfully captures the model’s reasoning: the visual contradiction of the color black is not ignored but is mathematically integrated to penalize the incorrect textual description.

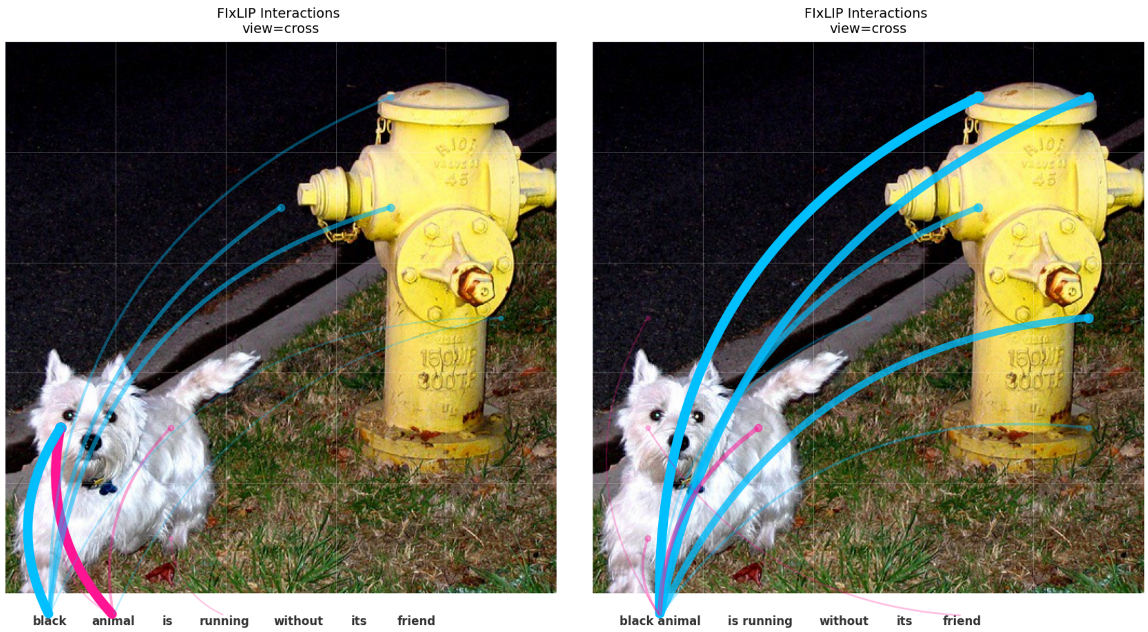

Figure 18: Interaction analysis for “black animal is running without its friend”

6.9 Visualization of Main Effects

Although our primary analysis focuses on cross-modal interactions, the FIxLIP framework naturally quantifies main effects — the standalone contribution of each image patch or text token. We visualize these effects by coloring the background of tokens and image grids: red indicates positive attribution (increasing similarity), while blue represents negative attribution.

Figure 19 revisits the "saddle embolus" example. In the `raw` strategy, not only are the connections fragmented, but the individual tokens (e.g., "sad", "dle") lack significant positive attribution (weak or neutral colors). In contrast, the `smart_depth` strategy reveals that the grouped concept "saddle embolus" carries a strong positive main effect, which is grounded in the central pulmonary region of the image.

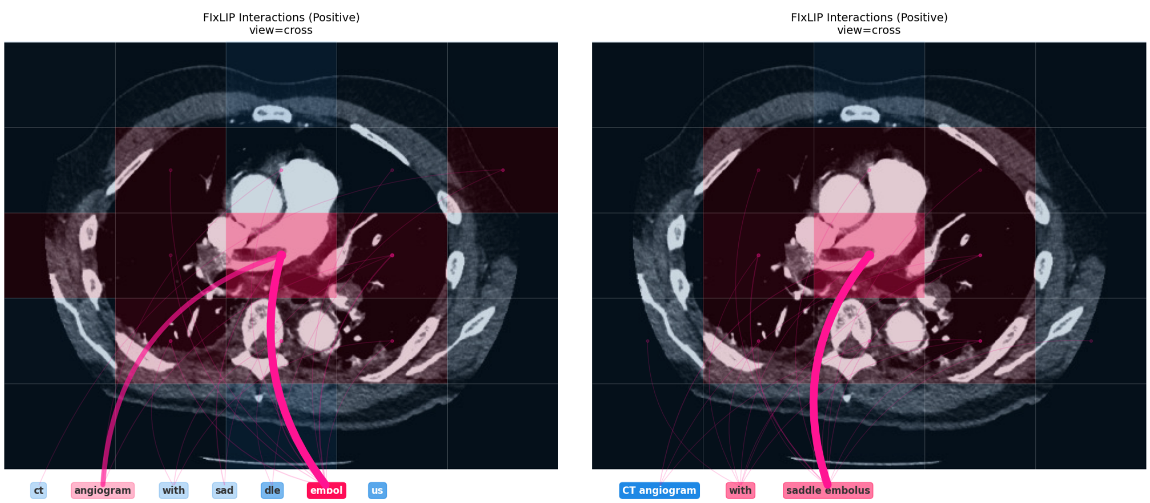

Figure 19: Main effects for "saddle embolus". **Left:** Fragmented tokens lack strong individual importance. **Right:** The grouped term becomes a dominant positive predictor, strongly grounded in the relevant anatomy.

A similar pattern is observed in the "Sagittal view of the calcified nasal packing" example in Figure 20. The `smart_depth` grouping allows the model to clearly attribute the "calcified nasal packing" to the specific hyperdense region and also "sagittal view" concept to the facial profile patches, while correctly assigning negative, neutral values to background noise, which is not possible with the noisy `raw` tokenization.

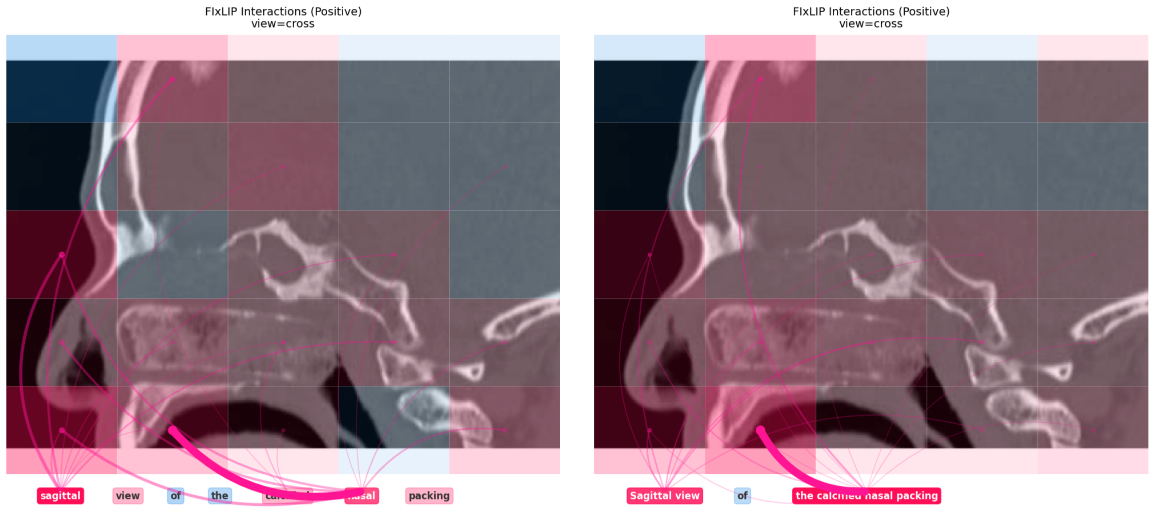

Figure 20: Main effects for "Sagittal view of the calcified nasal packing". **Left:** Noisy distribution of importance. **Right:** Clear semantic grounding where relevant image parts receive positive attribution.